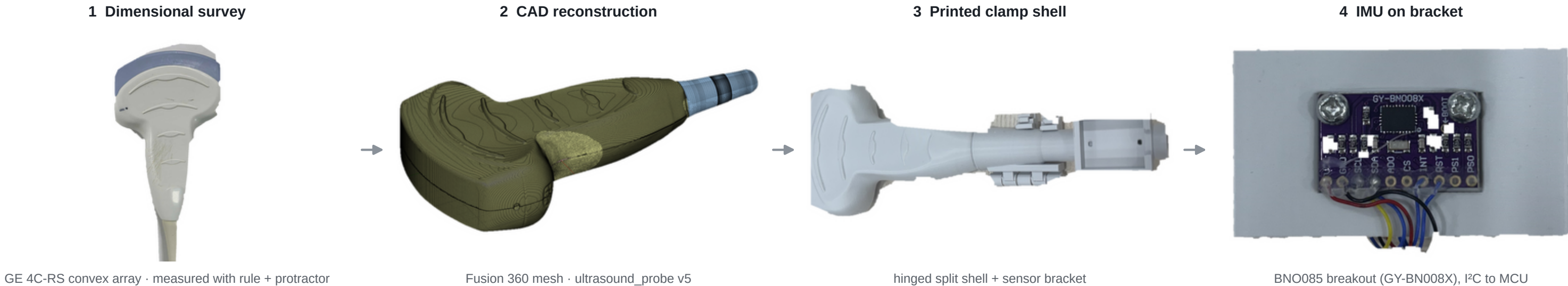
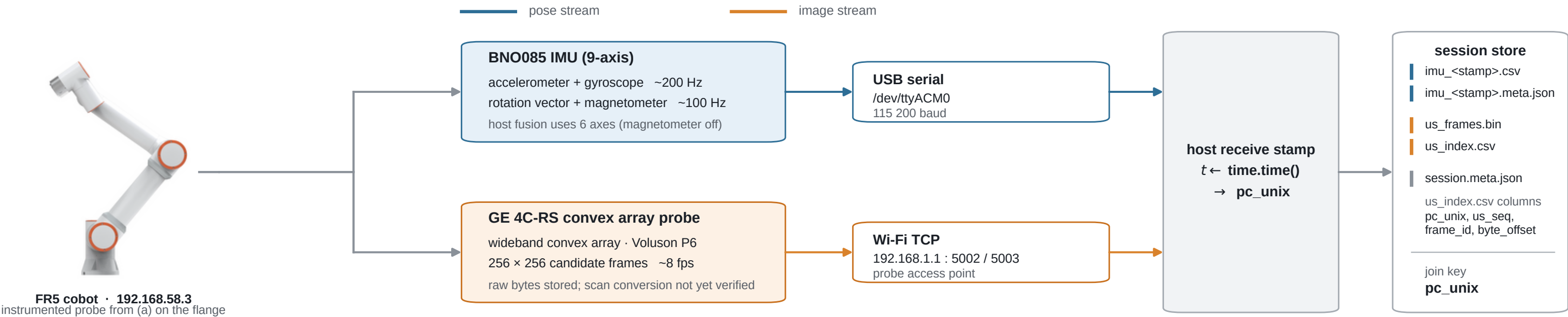


Sonologger — synchronized ultrasound and IMU acquisition

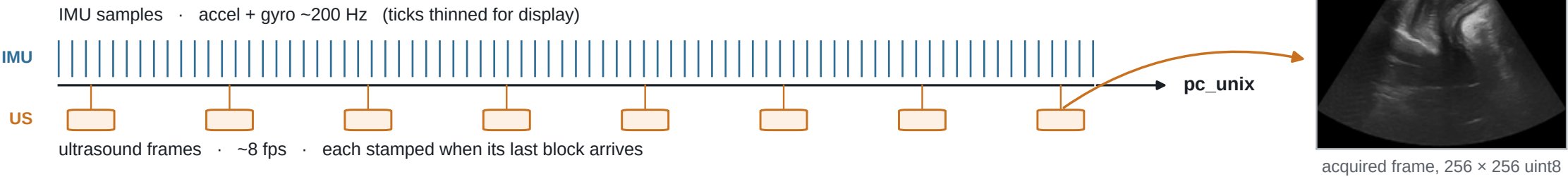
(a) Instrumented probe build



(b) Acquisition architecture — two paths, one clock



(c) Common time base



Alignment is by arrival time.

The IMU path is USB, microsecond-scale. The ultrasound path adds Wi-Fi TCP and in-probe acquisition delay. Clock skew is correctable afterwards; the fixed end-to-end ultrasound latency is not removed here — it remains to be measured.